

Appendix to "The Method of Moderation"

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Abstract

This appendix provides detailed mathematical derivations and technical results supporting the Method of Moderation. Topics include: value function transformations and their relationship to the inverse value function; explicit formulas for minimal and maximal marginal propensities to consume; cusp point calculations for tighter upper bounds; Hermite interpolation slope formulas and MPC derivations; patience conditions ensuring well-defined solutions; and extensions to stochastic returns with explicit formulas for portfolio problems.

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1. Appendix: Mathematical Details

1.1. Patience Conditions Details

Each patience condition from the main text controls a distinct way the problem could misbehave. The FVAC $0 < \beta G^{1-\rho} \mathbb{E}[\psi^{1-\rho}] < 1$ guarantees that even autarky, saving nothing and consuming income as it arrives, delivers finite expected discounted utility, so the consumer has a reason to value resources at all. The AIC $\Phi < 1$ rules out indefinite deferral of consumption: under certainty the marginal utility of consuming now exceeds the discounted marginal utility of consuming later. Two further conditions bound wealth from opposite directions. The RIC $\Phi/R < 1$ holds asset growth below the patience-adjusted discount rate, so wealth cannot explode; the GIC $\Phi/G < 1$ holds consumption growth below permanent-income growth, which is what pins down a finite target wealth ratio. Finally, the FHCW $G/R < 1$ keeps the present value of future labor income

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finite. Where these conditions fail, behavior changes qualitatively: when all hold, the consumer runs a buffer stock around a target wealth ratio; when the AIC fails, she borrows without limit; when the GIC fails but the RIC still holds, wealth grows without bound (Carroll, 1997, 2020; Carroll and Shanker, 2024).

1.2. Human Wealth Formulas

The optimist's human wealth (assuming $\xi_{t+n} = 1 \forall n > 0$) can be computed three ways: backward recursion $\bar{h}_T = 0$, $\bar{h}_t = (G/R)(1 + \bar{h}_{t+1})$; forward sum $\bar{h}_t = \sum_{n=1}^{T-t} (G/R)^n$; or infinite-horizon $\bar{h} = G/(R - G)$ when $R > G$. With $G = 1$, $\bar{h} = 1/(R - 1)$.

The pessimist's human wealth (assuming $\xi_{t+n} = \underline{\xi} \forall n > 0$) follows similarly: backward recursion $\underline{h}_T = 0$, $\underline{h}_t = (G/R)(\underline{\xi} + \underline{h}_{t+1})$; forward sum $\underline{h}_t = \underline{\xi} \sum_{n=1}^{T-t} (G/R)^n$; or infinite-horizon $\underline{h} = \underline{\xi} G/(R - G)$. When $\underline{\xi} = 0$ (unemployment), $\underline{h} = 0$.

1.3. Marginal Propensity to Consume Formulas

The minimal MPC (perfect foresight consumer with horizon $T - t$) has three forms (Carroll, 2009): backward recursion $\underline{\kappa}_t = \underline{\kappa}_{t+1}/(\underline{\kappa}_{t+1} + \Phi/R)$ with $\underline{\kappa}_T = 1$; forward sum $\underline{\kappa}_t = (\sum_{n=0}^{T-t} (\Phi/R)^n)^{-1}$; or infinite-horizon $\underline{\kappa} = 1 - \Phi/R = 1 - (R\beta)^{1/\rho}/R$.

The maximal MPC (Carroll and Toche, 2009) satisfies backward recursion $\bar{\kappa}_t = \bar{\kappa}_{t+1}/(\bar{\kappa}_{t+1} + \wp^{1/\rho}\Phi/R)$ with $\bar{\kappa}_T = 1$; forward sum $\bar{\kappa}_t = (\sum_{n=0}^{T-t} (\wp^{1/\rho}\Phi/R)^n)^{-1}$; or infinite-horizon $\bar{\kappa} = 1 - \wp^{1/\rho}(\Phi/R)$.

1.4. Cusp Point Calculation

The two upper bounds intersect at the cusp point m^* where

$$\begin{aligned} (\Delta m^* + \Delta h) \underline{\kappa} &= \bar{\kappa} \Delta m^* \\ \Delta m^* &= \frac{\underline{\kappa} \Delta h}{\bar{\kappa} - \underline{\kappa}} \\ m^* &= -\underline{h} + \frac{\underline{\kappa} (\bar{h} - \underline{h})}{\bar{\kappa} - \underline{\kappa}}, \end{aligned} \tag{1}$$

where $\Delta m^* \equiv m^* - \underline{m} > 0$ since $\bar{\kappa} > \underline{\kappa}$. For $m \in (\underline{m}, m^*]$, the tighter upper bound yields

$$\begin{aligned} \Delta m \underline{\kappa} &< \hat{c}(\underline{m} + \Delta m) &< \bar{\kappa} \Delta m \\ 0 &< \hat{c}(\underline{m} + \Delta m) - \Delta m \underline{\kappa} &< \Delta m (\bar{\kappa} - \underline{\kappa}) \\ 0 &< \left(\frac{\hat{c}(\underline{m} + \Delta m) - \Delta m \underline{\kappa}}{\Delta m (\bar{\kappa} - \underline{\kappa})} \right) &< 1. \end{aligned} \tag{2}$$

This motivates the definition of the low-resource moderation ratio as in (??).

1.5. Value Function Derivation

Under perfect foresight, consumption grows at constant rate equal to the absolute patience factor Φ : $\mathbf{c}_{t+n} = \mathbf{c}_t \Phi^n$. The present discounted value of consumption satisfies $\text{PDV}_t^T(\mathbf{c}) = \sum_{n=0}^{T-t} \beta^n \mathbf{c}_t \Phi^n = \mathbf{c}_t \sum_{n=0}^{T-t} (\Phi/R)^n$, where we use $\beta \Phi^{1-\rho} = \Phi/R$. Dividing by consumption yields the PDV-to-consumption ratio $\mathcal{C}_t^T = \text{PDV}_t^T(\mathbf{c})/\mathbf{c}_t = \sum_{n=0}^{T-t} (\Phi/R)^n = \underline{\kappa}_t^{-1}$, which is unchanged for normalized variables. Defining $\mathcal{C} \equiv \lim_{T \rightarrow \infty} \mathcal{C}_t^T$, this yields the key identity $\mathcal{C} = \underline{\kappa}^{-1}$, connecting the infinite-horizon PDV-to-consumption ratio to the minimal MPC.

The optimist's value function satisfies

$$\begin{aligned} \bar{\mathbf{v}}_{T-1}(m_{T-1}) &\equiv \mathbf{u}(c_{T-1}) + \beta \mathbf{u}(c_T) \\ &= \mathbf{u}(c_{T-1}) (1 + \beta \Phi^{1-\rho}) \\ &= \mathbf{u}(c_{T-1}) (1 + \Phi/R) \\ &= \mathbf{u}(c_{T-1}) \mathcal{C}_{T-1}^T \end{aligned} \tag{3}$$

The infinite horizon expression becomes

$$\begin{aligned} \bar{\mathbf{v}}(m) &= \mathbf{u}(\bar{\mathbf{c}}(m)) \mathcal{C} \\ &= \mathbf{u}(\bar{\mathbf{c}}(m)) \underline{\kappa}^{-1} \\ &= \mathbf{u}((\Delta m + \Delta h) \underline{\kappa}) \underline{\kappa}^{-1} \\ &= \mathbf{u}(\Delta m + \Delta h) \underline{\kappa}^{-\rho}. \end{aligned} \tag{4}$$

This can be transformed as

$$\begin{aligned} \bar{\Lambda} &\equiv ((1 - \rho) \bar{\mathbf{v}})^{1/(1-\rho)} \\ &= c \mathcal{C}^{1/(1-\rho)} \\ &= (\Delta m + \Delta h) \underline{\kappa}^{-\rho/(1-\rho)}. \end{aligned} \tag{5}$$

For the realist's problem, we define $\hat{\Lambda} = ((1 - \rho) \hat{\mathbf{v}}(m))^{1/(1-\rho)}$. Using the bounds $\underline{\Lambda} < \hat{\Lambda} < \bar{\Lambda}$, we define

$$\hat{\Omega}(\mu) = \left(\frac{\hat{\Lambda}(\underline{m} + e^\mu) - \underline{\Lambda}(\underline{m} + e^\mu)}{\Delta h \underline{\kappa} \mathcal{C}^{1/(1-\rho)}} \right) \tag{6}$$

and the logit-transformed counterpart:

$$\begin{aligned} \hat{\mathbf{X}}(\mu) &= \log \left(\frac{\hat{\Omega}(\mu)}{1 - \hat{\Omega}(\mu)} \right) \\ &= \log(\hat{\Omega}(\mu)) - \log(1 - \hat{\Omega}(\mu)) \end{aligned} \tag{7}$$

Inverting these approximations yields

$$\hat{\Lambda} = \underline{\Lambda} + \overbrace{\left(\frac{1}{1 + \exp(-\hat{\mathbf{X}})} \right)}^{=\hat{\Omega}} \Delta h \underline{\kappa} \mathcal{C}^{1/(1-\rho)} \tag{8}$$

from which the value function approximation is $\hat{\mathbf{v}} = \mathbf{u}(\hat{\Lambda})$.

1.6. I.I.D. Stochastic Returns: MPC derivation

The fact that a linear consumption function with an $\text{MPC} = 1 - (\beta \mathbb{E}[\mathbf{R}^{1-\rho}])^{1/\rho}$ satisfies the Euler equation with i.i.d. returns and no labor income can be derived by the method of undetermined coefficients. In particular, assume that $\bar{c}(m) = m\kappa$, with a time-independent MPC κ to be determined. Substituting this into the Euler equation, we have

$$\begin{aligned} 1 &= \beta \mathbb{E}_t \left[\mathbf{R}_{t+1} \left(\frac{c_{t+1}}{c_t} \right)^{-\rho} \right] \\ &= \beta \mathbb{E}_t \left[\mathbf{R}_{t+1} \left(\frac{m_{t+1}}{m_t} \right)^{-\rho} \right] \end{aligned} \quad (9)$$

where the second equality uses the assumed form of the consumption function. Since there is no labor income, $m_{t+1} = \mathbf{R}_{t+1}(m_t - c_t)$. Substituting this into the above we obtain

$$1 = \beta \mathbb{E}_t \left[\mathbf{R}_{t+1} (\mathbf{R}_{t+1}(1 - \kappa))^{-\rho} \right] \quad (10)$$

Solving for κ and recalling that returns are i.i.d. gives $\kappa = 1 - (\beta \mathbb{E}[\mathbf{R}^{1-\rho}])^{1/\rho}$.

In the particular case of lognormal returns, the MPC can be written in closed form. The moment generating function (MGF) for lognormal returns provides the key formula. For $\log \mathbf{R} \sim \mathcal{N}(\mu, \sigma^2)$, the MGF is $\mathbb{E}[e^{sX}] = \exp(\mu s + \sigma^2 s^2/2)$ where $X = \log \mathbf{R}$. Setting $s = 1 - \rho$ and $\mu = r + \pi - \sigma_{\mathbf{r}}^2/2$ yields¹

$$\mathbb{E}[\mathbf{R}^{1-\rho}] = \exp \left((1 - \rho) \left(r + \pi - \frac{\sigma_{\mathbf{r}}^2}{2} \right) + \frac{(1 - \rho)^2 \sigma_{\mathbf{r}}^2}{2} \right). \quad (11)$$

Simplifying the variance terms: $(1 - \rho)^2 \sigma_{\mathbf{r}}^2/2 - (1 - \rho) \sigma_{\mathbf{r}}^2/2 = (1 - \rho)[(1 - \rho) - 1] \sigma_{\mathbf{r}}^2/2 = -\rho(1 - \rho) \sigma_{\mathbf{r}}^2/2$, giving the final form

$$\mathbb{E}[\mathbf{R}^{1-\rho}] = \exp \left((1 - \rho) (r + \pi - \rho \sigma_{\mathbf{r}}^2/2) \right). \quad (12)$$

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¹Here we can interpret π as the risk premium, that is, the additional average return from holding a risky asset compared to the risk-free rate r . Adjusting the average log return by the asset volatility ensures that increasing $\sigma_{\mathbf{r}}^2$ constitutes a mean-preserving spread of the level of return.

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